

Self-healing and self-renewing low surface energy fluorescent antifouling coatings based on dynamic reversible covalent bonds

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ABSTRACT

To address the environmental toxicity concerns associated with traditional organotin antifouling coatings, this study proposes a novel eco-friendly antifouling coating based on multifunctional synergistic mechanisms. A dynamic imine bond-mediated crosslinking strategy was employed to integrate aminopropyl-terminated polydimethylsiloxane (PDMS) with aldehydic fluorescent molecules (BTB, TPB, SPB), creating a multifunctional coating that combines low surface energy, self-healing properties, fluorescent antifouling capabilities, and pH-responsive self-renewal characteristics. Experimental results demonstrate that the coating achieves dynamic optimization of surface free energy (19.8–24.6 mJ/m²) through PDMS molecular weight regulation. The imine bonds endow the material with exceptional self-healing performance (complete recovery within 48 h) and acid-responsive degradation behavior. Fluorescence synergy significantly enhances antifouling efficiency, with blue-light-emitting coatings achieving a 99.84 % antibacterial rate against *Escherichia coli*, while red-light-emitting coatings reduce diatom adhesion by 71.49 % under illumination. The PDMS-based coating with a molecular weight of 3000 demonstrates optimal comprehensive performance, where the balance between fracture strength (0.37 MPa) and self-renewal rate achieves synergistic enhancement of static antifouling and dynamic responsiveness. This research provides a new strategy for developing environmentally intelligent antifouling coatings, overcoming the performance limitations of single-mechanism antifouling technologies through multi-mechanism synergistic effects.

1. Introduction

Marine vessels face persistent biofouling challenges during navigation and docking, where marine organisms adhere to hull surfaces [1]. Despite their small individual size, these organisms exhibit vast diversity and large-scale colonization, increasing vessel weight, hydrodynamic resistance, and fuel consumption, thereby elevating maintenance costs and reducing economic efficiency [2]. While traditional organotin antifouling coatings effectively eliminate fouling organisms, their potent toxicity enables bioaccumulation and causes irreversible environmental harm [3]. Consequently, the development of environmentally benign alternatives has become imperative.

In the absence of toxic additives, "non-stick" coatings based on weak interfacial adhesion have emerged as a promising strategy [4]. However, achieving effective antifouling through a single mechanism remains challenging due to the diversity of fouling organisms and their varied adhesion mechanisms [5]. A synergistic integration of multiple

antifouling strategies has proven more efficient [6], including low surface free energy [7], eco-friendly antifoulant incorporation [8], self-renewal [9], and fluorescent antifouling [10]. Recent advances highlight biomimetic antifouling approaches for their environmental compatibility. Low surface energy coatings inhibit bioadhesion by minimizing surface free energy, yet their static antifouling mechanisms are prone to failure due to physical damage or chemical aging [7]. The introduction of self-healing materials offers a novel solution [11,12], where dynamic covalent bonds [13] (e.g., Diels-Alder reactions) or supramolecular interactions [14] (e.g., hydrogen bonds, coordination bonds) enable autonomous repair of mechanical damage, extending coating durability [15].

Nevertheless, single self-healing mechanisms struggle to address the dynamic complexity of marine environments, necessitating synergistic multifunctional enhancements [16]. For instance, dynamic surface self-renewal can continuously expose fresh antifouling interfaces through controlled degradation or phase separation. Fluorescent

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antifouling systems, utilizing materials such as long-afterglow phosphors [17], rare-earth complexes [18], and organic fluorophores [19], not only suppress microbial adhesion via photodynamic effects [20] but also enable real-time, in situ monitoring of coating integrity [21].

Light intensity and wavelength significantly influence the motility, adhesion, and proliferation behaviors of marine fouling organisms [22]. These effects are particularly pronounced in single-celled organisms such as diatoms [23], making light modulation a promising strategy for biofouling inhibition [24]. Jin et al. proposed that fluorescent corals emit multicolored weak fluorescence under blue and violet light, which deters fouling organisms from adhering to their surfaces. Inspired by this mechanism, they designed a sandwich-structured coating comprising polydimethylsiloxane (PDMS), phosphor, and silicone rubber [25]. This coating effectively suppresses algal attachment and reduces bacterial adhesion through fluorescence. Gou et al. developed an antifouling coating by grafting 7-amino-4-methylcoumarin—a compound with both fluorescent and bactericidal properties—onto PDMS and poly-tetrahydrofuran chains, achieving synergistic antifouling through fluorescence and contact-based antibacterial mechanisms [19]. The coating demonstrated remarkable antifouling performance under illumination. Similarly, Hou et al. fabricated a zinc acrylate polymer grafted with coumarin, integrating fluorescence-driven antifouling, natural bactericidal activity, and self-polishing surface renewal into a multifunctional system [26]. These studies highlight composite coatings where fluorescent antifouling serves as the primary strategy, augmented by complementary mechanisms. The fluorescence primarily originates from long-afterglow phosphors or sidechain-grafted fluorophores.

This study proposes a novel multifunctional synergistic antifouling coating design: (1) A dynamically crosslinked network constructed via reversible covalent bonds enables self-healing and stimuli-responsive self-renewal; (2) A PDMS-based low-surface-energy substrate minimizes initial biofouling adhesion; (3) Aldehyde-functionalized organic fluorophores, serving as crosslinking nodes for silicone chains, synergistically achieve fluorescence-mediated microbial repulsion and sterilization, inhibiting early-stage microbial colonization. Specifically, aminopropyl-terminated PDMS was crosslinked with aldehyde-functionalized benzothiadiazole, tetraphenylethylene, and spirobi-fluorene via Schiff base reactions, forming dynamic imine bonds to impart self-healing properties. These acid-labile imine bonds undergo hydrolysis under the slightly acidic conditions generated by bacterial membranes [27], enabling controlled self-renewal through structural disassembly. Concurrently, sunlight stimulation triggers fluorescence emission at specific wavelengths from the fluorophores [20], achieving microbial repulsion. The synergistic integration of these three antifouling mechanisms significantly enhances coating performance, offering a novel solution for marine antifouling applications.

2. Material and methods

2.1. Materials

The silicone matrix employed was amine-terminated polydimethylsiloxane (PDMS) with number-average molecular weights ($M_n = 3000, 5000, 7000, \text{ and } 10000, \geq 98\%$). The crosslinker benzene-1,3,5-tricarbaldehyde ($\geq 98\%$) was used. All the above materials were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. The fluorophores included 2,2,7,7-tetrabromo-9,9'-spirobi[fluorene] ($\geq 99\%$), tetrabromotetraphenylethylene ($\geq 98\%$), and 4,7-bis(5-bromo-2-thienyl)-2,1,3-benzothiadiazole ($\geq 98\%$). The catalysts and reagents comprised tetrakis(triphenylphosphine)palladium ($\geq 98\%$), 4-formylphenylboronic acid pinacol ester ($\geq 97\%$). All the above materials were purchased from Shanghai Aladdin Biochemical Technology Co., Ltd. potassium carbonate (AR), 1,4-dioxane (AR), dichloromethane (AR), anhydrous sodium sulfate (AR), and tetrahydrofuran (AR). All the above materials were purchased from Shanghai Macklin Biochemical Technology Co., Ltd.

2.2. Synthesis of aldehyde-functionalized fluorophores

The synthetic routes and protocols for three aldehyde-functionalized fluorophores with distinct emission wavelengths are detailed in the **Supplementary Materials**.

2.3. Coating fabrication

Transparent self-healing fluorescent coatings were prepared by mixing 0.8 g of diamino-propyl-terminated PDMS ($M_n = 3000, 5000, 7000 \text{ and } 10000$) with 0.4 mL of 0.26 M benzene-1,3,5-tricarbaldehyde and 2.5×10^{-5} M fluorophores (BTB, TPB, or SPB) in THF. The mixture was cast into a polytetrafluoroethylene (PTFE) mold ($51 \text{ mm} \times 77 \text{ mm} \times 5 \text{ mm}$) and dried at room temperature for 48 h to facilitate imine bond formation and PDMS crosslinking. After demolding, irregular edges were trimmed. For tensile testing, dumbbell-shaped specimens ($75 \text{ mm} \times 4 \text{ mm} \times 2 \text{ mm}$, effective span: 30 mm) were cut. Optical testing samples were prepared as $30 \text{ mm} \times 30 \text{ mm} \times 2 \text{ mm}$ plates. For antifouling and surface characterization, coatings were fabricated on glass slides ($25.4 \text{ mm} \times 76.2 \text{ mm} \times 1 \text{ mm}$) with a thickness of 1 mm.

2.4. Characterization

2.4.1. Surface morphology

Coating surface morphology was analyzed using a confocal laser scanning microscope (CLSM; Olympus OLS4000).

2.4.2. Surface wettability and free energy

Contact angles were measured via the sessile drop method using a contact angle goniometer (JC2000C). Droplets (3 μL) of diiodomethane and deionized water were deposited on the coatings. Prior to testing, samples were ultrasonically cleaned in anhydrous ethanol for 5 min, rinsed with deionized water for 5 min, and dried. Surface free energy (SFE) was calculated using the Owens two-liquid method [28], according to Eqs. (1), (2) and (3), with results averaged from six parallel measurements. The corresponding parameters are defined as follows: $\theta_{\text{H}_2\text{O}}$ represents the water contact angle, $\theta_{\text{CH}_2\text{I}_2}$ denotes the diiodomethane contact angle, σ_s^p is the polar component, σ_s^d is the dispersion component, and σ_s stands for the surface free energy, with units of mJ/m^2 .

$$\sigma_s^p = [(137.5 + 256.1 \times \cos\theta_{\text{H}_2\text{O}} - 118.6 \times \cos\theta_{\text{CH}_2\text{I}_2})/44.92]^2 \quad (1)$$

$$\sigma_s^d = [(139.9 + 181.4 \times \cos\theta_{\text{CH}_2\text{I}_2} - 41.5 \times \cos\theta_{\text{H}_2\text{O}})/44.92]^2 \quad (2)$$

$$\sigma_s = \sigma_s^p + \sigma_s^d \quad (3)$$

2.4.3. Tensile testing of coatings

The mechanical properties of the coatings were examined using an electronic universal testing machine (Wance UTM-5105) equipped with microprocessor control. Dumbbell-shaped tensile specimens ($75 \text{ mm} \times 4 \text{ mm}$) were cut from the coatings prepared in PTFE molds. The initial gauge length was set to 25 mm, and the tensile speed was maintained at 100 mm/min. Each coating was tested in three parallel replicates. Stress-strain curves, elongation at break, and fracture strength of the coatings were determined. To approximate the Young's modulus, the most linear segment of the stress-strain curve was selected for linear fitting, with the slope of the fitted line representing the modulus. The self-healing efficiency could be obtained by calculating the ratio of the tensile strength of the healed samples to that of the original samples.

2.4.4. UV-Vis spectroscopy of coatings

Ultraviolet-visible (UV-Vis) spectra of the coatings were acquired using a UV-Vis spectrophotometer (DRS, UV-3600 Plus). Coating films with a thickness of 150 μm were prepared for analysis. Measurements were performed in high-speed scan mode with air as the reference

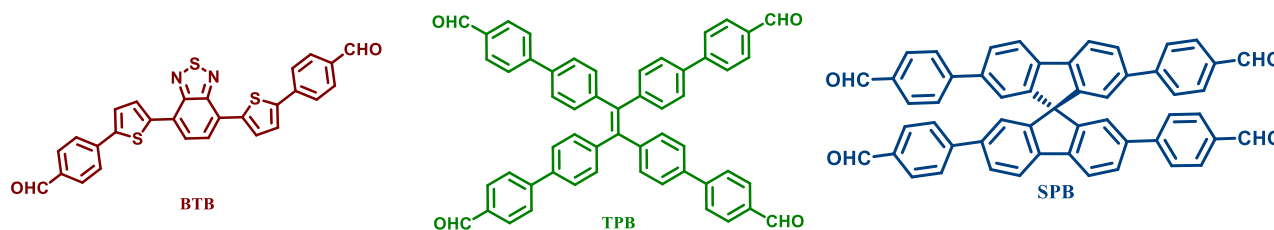


Fig. 1. Structural formulas of three fluorescent molecules.

background, covering a wavelength range of 190–900 nm.

2.4.5. Fluorescence testing of coatings

Photoluminescence (PL) spectra and excitation spectra of the samples were obtained using a fluorescence spectrophotometer (Hitachi F4600). The excitation wavelength for each sample was determined based on its excitation spectrum.

2.5. Antibacterial performance of coatings

The antibacterial efficacy of the coatings against *Escherichia coli* (*E. coli*) and *Staphylococcus aureus* (*S. aureus*) was assessed using the plate counting method. All experimental apparatuses were sterilized by autoclaving at 121 °C. *E. coli* and *S. aureus* were cultured in a constant-temperature incubator at 37 °C for 24 h. Activated bacteria were inoculated into Luria-Bertani (LB) liquid medium using an inoculation loop and incubated for an additional 24 h. Bacterial suspensions were diluted to appropriate concentrations using a UV–Vis spectrophotometer. The cultured suspensions were diluted 100-fold with phosphate-buffered saline (PBS), and 40 mL of the diluted suspension was used to immerse specimens under fluorescent light irradiation for 24 h. After retrieval, specimens were rinsed three times with 10 mL PBS each and transferred to centrifuge tubes containing 40 mL PBS. Samples were sonicated for 15 min, and 100 µL of the supernatant was uniformly spread on LB agar plates using a spreader until complete absorption. The plates were incubated at 37 °C for 24 h. Bacterial colonies were enumerated post-incubation [5].

2.6. Anti-Diatom adhesion testing of coatings

Diatoms contain chlorophyll-a, and thus the extraction of chlorophyll-a from organisms on the coatings reflects the quantity of diatoms attached to the surface. *Navicula tenera* was selected as the model diatom for these experiments. Chlorophyll-a levels were used to evaluate the concentration of *Navicula tenera* adhering to the coatings, with lower chlorophyll concentrations indicating superior resistance of the coatings to diatom attachment. Two sets of coating specimens were immersed in plastic containers (20 cm × 10 cm × 5 cm) containing 500 mL of algal solution under continuous light at 22 °C for 24 h, while another set was kept in continuous darkness for the same duration. After immersion, the samples were rinsed with 30 mL of sterile deionized water. An extraction solution containing 45 mL of 90 % acetone and 1 mL of magnesium carbonate solution was prepared in centrifuge tubes. Defatted cotton swabs were dipped into the extraction solution and gently swabbed over the sample surfaces 5–6 times to collect adherent diatoms. The swabs were then soaked in the centrifuge tubes containing the extraction solution and stored in a biochemical incubator at 8 °C for 24 h to extract chlorophyll-a. Subsequently, 10 mL of the algal solution was aspirated with a pipette and centrifuged at 4000 r·min⁻¹ for 15 min. Finally, 3 mL of the supernatant was collected, and its absorbance was measured at wavelengths of 750, 663, 645, and 630 nm using a UV-2000 UV–Vis spectrophotometer. The parameter *S* (19.36 cm²), representing the area of the glass slide (25.4 mm × 76.2 mm × 1 mm), was used for normalization [29]. The corrected absorbance values, denoted as

OD₆₆₃, OD₆₄₅, and OD₆₃₀, were obtained by subtracting the absorbance at 750 nm from the absorbance values measured at 663, 645, and 630 nm, respectively. Thus, chlorophyll-a adhered on the coating was calculated using Eq. (4).

$$m_a = [11.64(OD_{663} - OD_{750}) - 2.16(OD_{645} - OD_{750}) + 0.10(OD_{630} - OD_{750})]/19.36 \quad (4)$$

2.7. Theoretical computational methods

Density Functional Theory (DFT) was employed to obtain precise computational results. Molecular models were constructed using GaussView 5 software, and theoretical calculations were performed with the Gaussian 09 program. Geometry optimizations and frequency calculations were conducted at the B3LYP/6–31G(d) level [30]. Frontier molecular orbital (FMO) information was extracted using Multiwfn 3.8 software [31], and molecular orbitals were visualized with VMD 1.9.4 software [32].

3. Results and discussions

3.1. Discussion on fluorescent monomer synthesis

Fig. 1 illustrates the structural formulas of the three fluorescent molecules, while Schemes S1–S3 outline the synthetic routes for the red-, green-, and blue-emitting aldehyde-containing fluorophores (BTB, TPB, and SPB), respectively. All three fluorescent compounds (red, green, and blue) were successfully synthesized via a one-pot Suzuki coupling reaction. Post-synthetic purification was achieved using Soxhlet extraction followed by column chromatography. The red-emitting compound BTB was obtained as a red solid powder with a final yield of 24 %. The green-emitting compound TPB formed a yellow-green solid powder with a yield of 51 %, and the blue-emitting compound SPB yielded a white solid powder at 45 %. Characterization data for the final products are provided in Figures S1–S3, confirming their successful synthesis. Notably, the red (BTB) and blue (SPB) fluorophores represent novel compounds unreported in the literature.

3.2. Photophysical analysis of fluorescent compounds

The three fluorescent molecules share similar structural frameworks, each consisting of a central chromophore conjugated with multiple aldehyde-substituted benzene rings. Their emission properties are primarily governed by the central chromophore, while the introduction of benzene rings on the side chains modulates their fluorescence characteristics. To elucidate this, the electronic energy levels of the fluorophores were simulated using Density Functional Theory (DFT) at the B3LYP/6–31G(d) level [33], enabling optimization of molecular conformations and calculation of HOMO, LUMO, and bandgap energies. As illustrated, BTB exhibits a bandgap of 2.45 eV, narrower than those of TPB and SPB. This reduction arises from the planar configuration of BTB, which enhances electron delocalization compared to the orthogonal geometries of TPB and SPB [34]. Furthermore, the sp³-hybridized central carbon atom in SPB induces a more orthogonal molecular

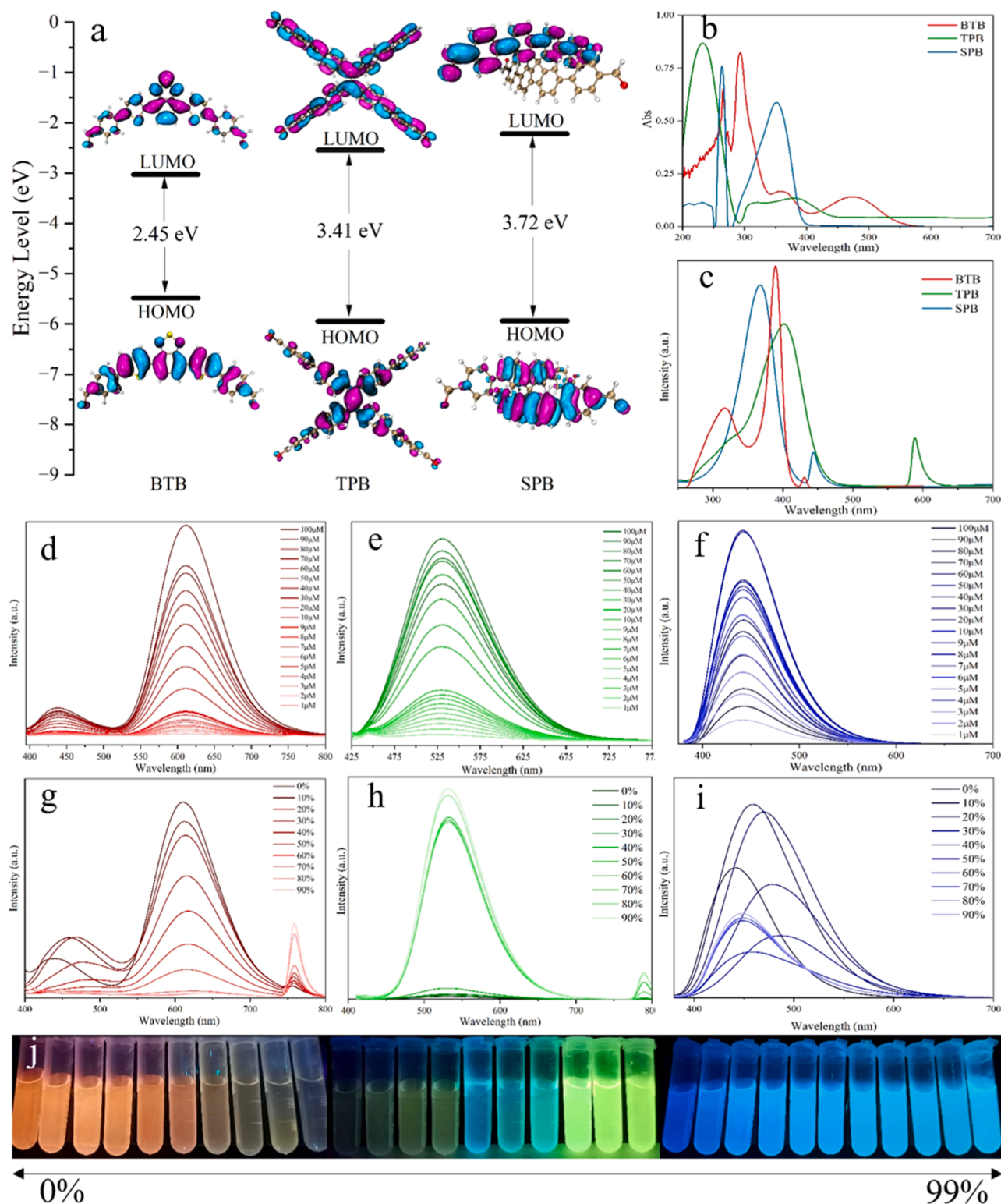


Fig. 2. (a) Energy level diagram of molecular orbital. (b) UV absorption spectrum. (c) Excitation spectra. (d) Concentration gradient fluorescence spectra of BTB in DMF. (e) Concentration gradient fluorescence spectra of TPB in DMF. (f) Concentration gradient fluorescence spectra of SPB in DMF. (g) Fluorescence spectra of BTB in THF-water. (h) Fluorescence spectra of TPB in THF-water. (i) Fluorescence spectra of SPB in THF-water. (j) Fluorescence of three fluorescent molecules in THF-water under a violet lamp.

conformation relative to TPB, leading to reduced electron delocalization and thus a larger bandgap (3.72 eV for SPB vs. 3.41 eV for TPB). The bandgap widens progressively from red- to blue-emitting compounds, attributed to enhanced conjugation through benzene ring incorporation. This increased conjugation accelerates relaxation during fluorescence

emission, reducing energy dissipation and narrowing emission spectra [33], consistent with the observed fluorescence profiles.

We systematically investigated the photophysical properties and aggregation-induced emission (AIE) behaviors of these fluorophores. BTB features a benzothiadiazole core linked to two thiophene units,

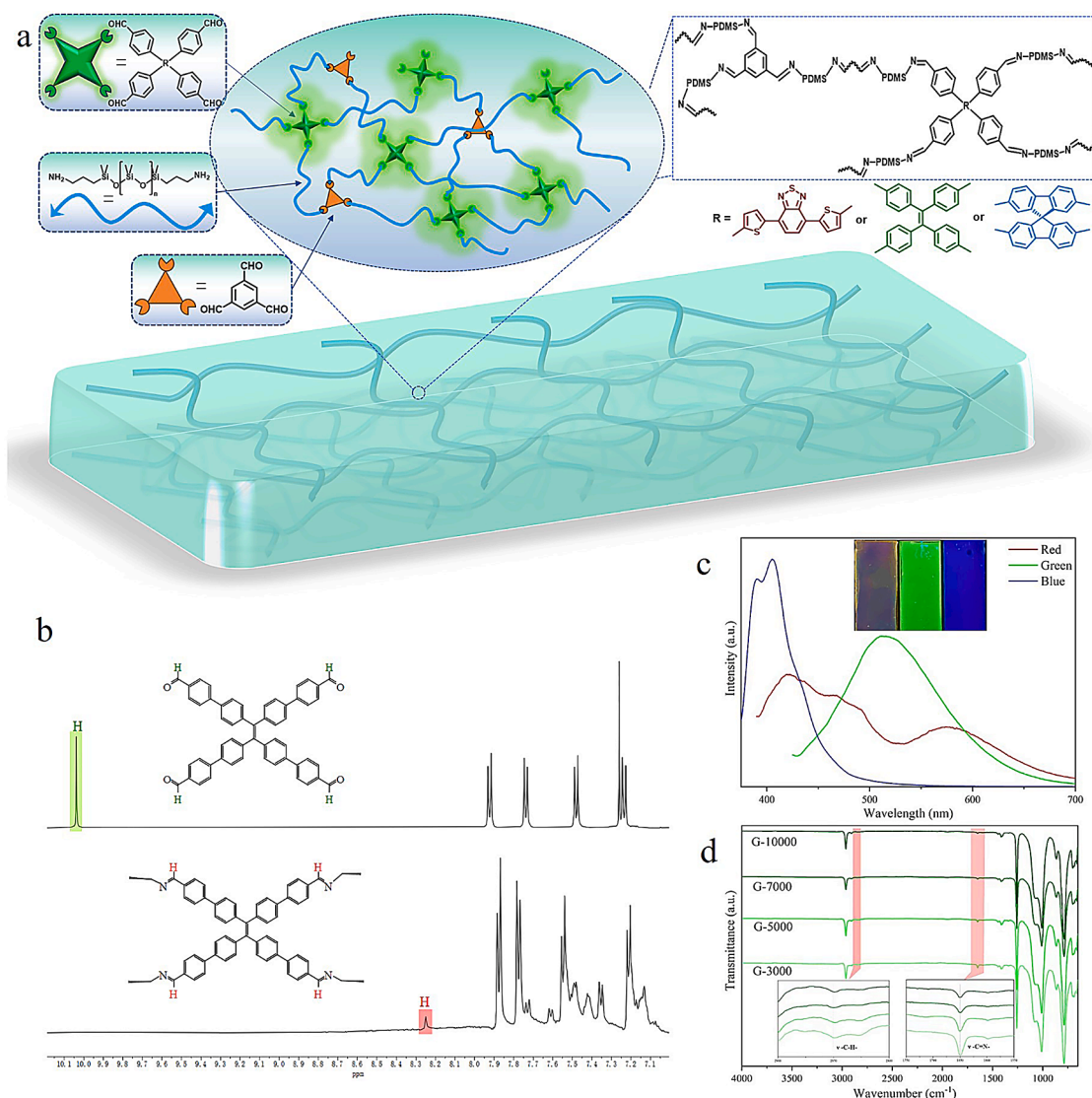


Fig. 3. (a) Schematic structure of fluorescent coating. (b) ^1H NMR Titration spectrum of TPB. (c) Fluorescence photos and fluorescence spectra of three coatings under UV lamp. (d) Infrared spectra of green fluorescent coatings.

enabling deep red emission in solution. As shown in Fig. 2b–d, its maximum absorption peak occurs at 300 nm, with excitation and emission maxima at 370 nm and 610 nm, respectively. Fluorescence intensity increases proportionally with concentration across tested gradients. TPB incorporates a tetraphenylethylene (TPE) core, a classic AIE-active motif, yielding yellow-green emission. Fig. 2b, c, e reveals its strongest absorption at 239 nm, with excitation and emission maxima at 405 nm and 530 nm, respectively. Similarly, fluorescence intensity escalates linearly with concentration. SPB employs a spirofluorene chromophore, a conventional blue-emitting unit, producing bright blue fluorescence. As depicted in Fig. 2b, c, f, SPB exhibits a primary absorption peak at 270 nm, with excitation and emission maxima at 370 nm and 440 nm. Fluorescence intensity increases with concentration from 1 to 20 μM but declines sharply at 30 μM , reaching a minimum at 100 μM .

3.2. Investigation of aggregation-induced emission (AIE) properties

To investigate the aggregation-induced emission (AIE) characteristics of the three fluorophores, varying proportions of a poor solvent (H_2O , f_w) were incrementally added to their tetrahydrofuran (THF) solutions. As shown in Fig. 2g, the fluorescence intensity of BTB gradually

diminished with increasing f_w , indicating the absence of AIE behavior. In contrast, TPB exhibited distinct AIE activity (Fig. 2h). At $f_w = 0\text{--}30\%$, its fluorescence color remained yellow (similar to that in DMF) with gradually intensifying emission. Between $f_w = 40\text{--}60\%$, the emission shifted to cyan-blue, accompanied by a sharp rise in intensity. At $f_w = 70\text{--}90\%$, the fluorescence transitioned to green, reaching maximum intensity, thereby confirming TPB as a robust AIE-active compound. For SPB (Fig. 2i), fluorescence intensity initially increased and then decreased at $f_w = 0\text{--}50\%$, followed by a gradual intensity recovery at $f_w = 60\text{--}90\%$. Concurrently, the emission color shifted from blue to cyan-green, accompanied by a red-shifted emission peak. This non-monotonic behavior suggests that SPB does not strictly conform to classical AIE mechanisms.

Despite their structural similarities, the three fluorophores display divergent emission traits, likely due to differences in their core architectures. The benzothiadiazole core of BTB adopts a planar nine-atom configuration. Although linked to freely rotating thiophene units, its extended π -conjugation facilitates detrimental π - π stacking and inter-system crossing upon aggregation, quenching fluorescence. TPB, featuring a tetraphenylethylene (TPE) core a canonical AIE motif shows enhanced AIE activity due to restricted intramolecular rotation (RIR) enforced by its appended phenyl rings. In contrast, SPB's

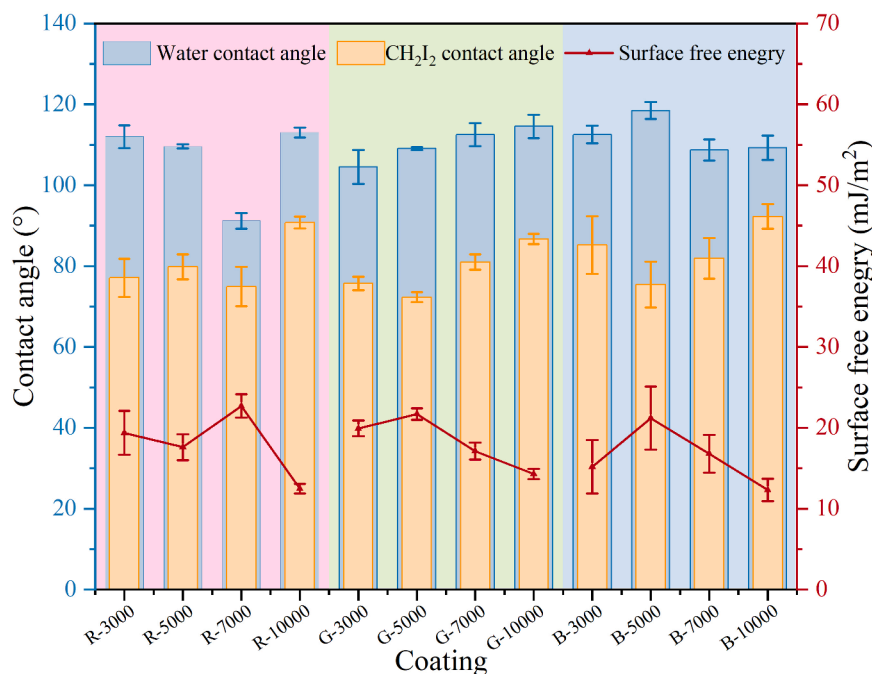


Fig. 4. Water contact angle, Diiodomethane contact angle and Surface energy of each coatings.

spiroconfigured fluorene rings impose high rigidity and orthogonality, effectively suppressing π - π stacking. However, the appended phenyl rings minimally influence its overall AIE behavior, resulting in non-classical emission trends.

3.3. Microstructure of fluorescent coatings

As illustrated in Fig. 3a, a self-healing fluorescent gel was prepared by mixing 2.4 g of bis(3-aminopropyl)-terminated polydimethylsiloxane (PDMS), 1.2 mL of 0.26 M triformylbenzene, and 2.5×10^{-5} M fluorophores in DMF. The mixture was cast into a polytetrafluoroethylene (PTFE) mold ($60 \times 60 \times 5$ mm) and dried at room temperature for two days, enabling imine bond formation and PDMS crosslinking. To confirm the Schiff base reaction between the aldehyde groups of the fluorophores and the terminal amine groups of PDMS, a nuclear magnetic resonance (NMR) titration experiment was conducted using TPB as an example (Fig. 3b). ^1H NMR Titration spectrum of BTB and SPB was shown in S4-S5. Upon adding excess amine-terminated PDMS to a deuterated dichloromethane solution of TPB, the proton signal of the aldehyde group at 10.03 ppm disappeared, replaced by a new signal at 8.25 ppm corresponding to the imine proton. This provides strong evidence for covalent bond formation between the aldehyde groups of TPB and the terminal amines of PDMS. The unchanged chemical shifts of PDMS segments before and after titration further confirm that imine bonds formed exclusively at the chain termini. This conclusion is corroborated by Fourier-transform infrared (FTIR) spectroscopy (Fig. 3d), where the characteristic imine bond absorption peak emerged at 1650 cm^{-1} . The intensity of this peak inversely correlated with monomer molecular weight, indicating a higher density of imine bonds in coatings derived from smaller monomers. Additionally, the aldehyde C-H stretching vibration at 2850 cm^{-1} exhibited intensity changes consistent with imine bond formation [35]. Red and blue fluorescent coatings are basically similar to green fluorescent coatings and show in S6-S7.

3.4. Fluorescent properties of the coatings

The aldehyde groups in BTB, TPB, and SPB reacted with the amine groups of PDMS to form crosslinked fluorescent coatings, endowing

them with distinct luminescent characteristics. As shown in Fig. 3c, the three coatings emitted red, green, and blue fluorescence under UV light, with emission maxima at 605, 530, and 440 nm, respectively. Despite the inherent aggregation-caused quenching (ACQ) tendency of the BTB fluorophore due to π - π stacking in aggregated states, the PDMS network effectively suppressed such interactions in the gel matrix. This confinement enabled intense red fluorescence even at low concentrations.

3.5. Surface properties of fluorescent coatings

The silicone backbone exhibits high flexibility, with significantly weaker intermolecular forces compared to hydrocarbons, resulting in lower viscosity, reduced surface tension, and diminished surface energy for silicone-based materials of equivalent molecular weight. These attributes—low surface tension and energy—underpin their broad applicability, including hydrophobicity, anti-adhesion, lubrication, and gloss enhancement. To evaluate surface characteristics, amine-terminated PDMS with molecular weights of 3000, 5000, 7000, and 10000 (denoted by color initials followed by molecular weight) was utilized. The water and diiodomethane contact angles of the coatings were measured, and their surface free energies were calculated (Fig. 4). As the molecular weight of the silicone monomers increased, the water contact angles exhibited a gradual upward trend. This behavior stems from the competing influences of silicone chain hydrophobicity and the polarity of dynamic covalent bonds within the coating. Siloxane segments act as hydrophobic domains, while dynamic covalent bonds introduce hydrophilic moieties. Higher molecular weights reduce the density of dynamic covalent bonds, thereby enhancing overall hydrophobicity. Concurrently, the calculated surface free energy progressively decreased, with the rate of reduction accelerating at higher molecular weights. This trend highlights the growing dominance of siloxane-derived hydrophobicity in governing surface energetics.

3.6. Self-Healing and mechanical properties of fluorescent coatings

The imine bonds formed in the polymer network are reversible dynamic covalent bonds, endowing the material with intrinsic self-healing capabilities. Taking the green-emitting coating (G-3000) as an example,

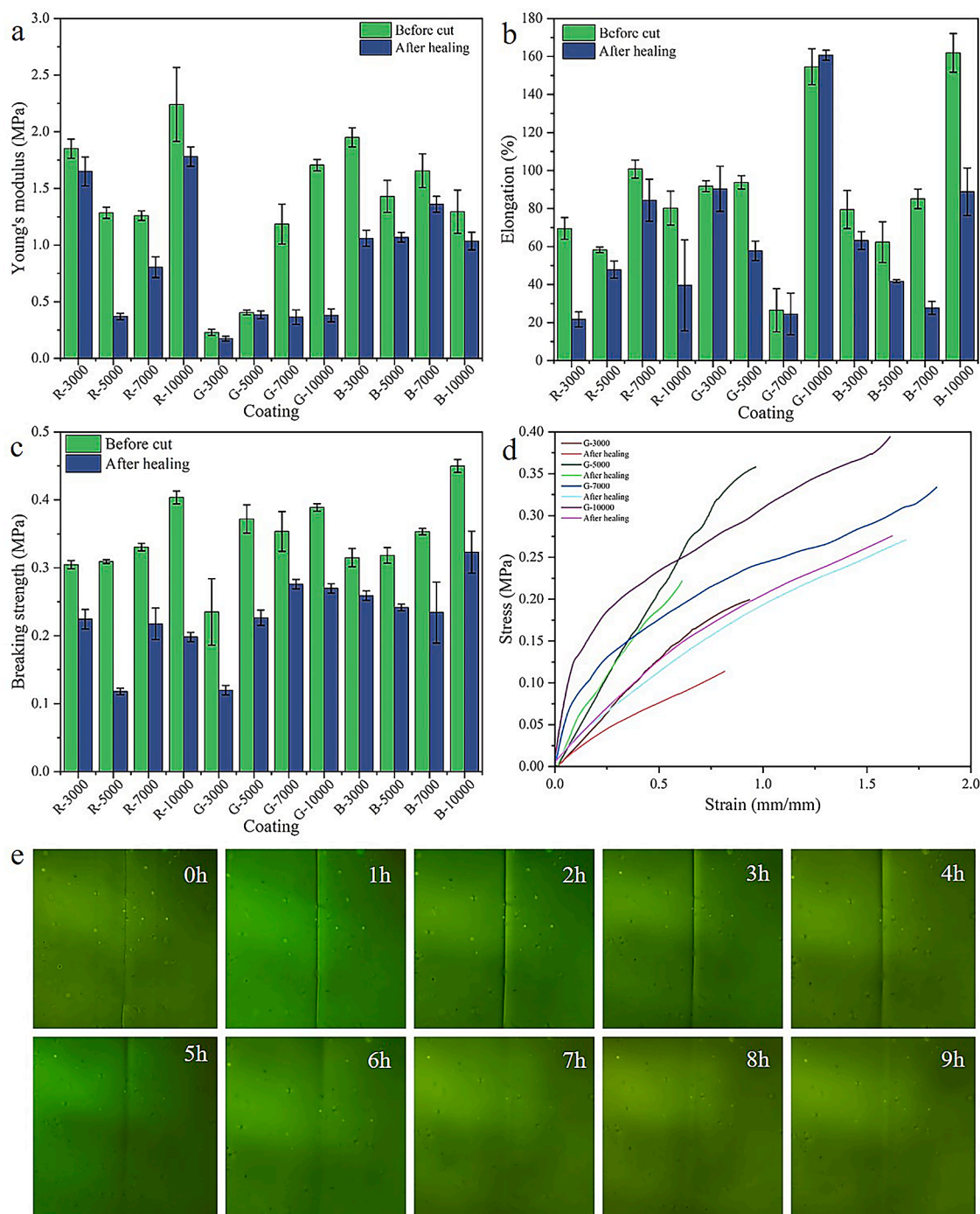


Fig. 5. Green coating Before cutting and after healing: (a) Young's modulus. (b) Elongation at break. (c) Breaking strength. (d) Stress-strain curves. (e) Photomicrograph showing the progression of wound healing over time.

a visible incision gradually diminished over 9 h of healing, indicating active self-repair (Fig. 5a). Similar behaviors were observed for G-5000, G-7000, and G-10000. As shown in Fig. 5a, incisions began to flatten within ~2 h post-cutting, followed by progressive healing over the next 4 h, with the scar becoming indistinct. Complete healing was achieved after 48 h.

Prior to incision, tensile tests were performed on coatings with varying monomer molecular weights. Higher molecular weights correlated with enhanced mechanical performance (Fig. 5a). Stress-strain curves revealed a progressive increase in Young's modulus for G-3000, G-5000, G-7000, and G-10000, with higher moduli indicating superior mechanical integrity. Notably, G-5000 exhibited nearly double the

fracture strength of G-3000, yet lower elongation at break compared to G-7000. This anomaly likely stems from variations in dynamic covalent bond density. As demonstrated in Fig. 3d, the FTIR absorption peak intensity for imine bonds ($\sim 1650\text{ cm}^{-1}$) decreased with increasing PDMS molecular weight, confirming fewer imine bonds in higher molecular weight networks. Larger monomers contain fewer terminal amine groups per unit mass, resulting in reduced dynamic crosslinking. Consequently, mechanical properties are governed by a balance between intrinsic chain strength (scaling with molecular weight) and dynamic covalent bond density. As shown in Fig. 5c, fracture strength peaked at a molecular weight of 5000, while elongation improved significantly at 7000. The stress-strain curves of the red and blue

Table 1
Healing efficiencies of the coatings.

Coating	Self-healing efficiency (%)	Coating	Self-healing efficiency (%)	Coating	Self-healing efficiency (%)
R-3000	73.39	G-3000	50.93	B-3000	82.28
R-5000	38.11	G-5000	60.94	B-5000	75.90
R-7000	65.83	G-7000	78.07	B-7000	66.27
R-10000	49.12	G-10000	69.36	B-10000	71.78

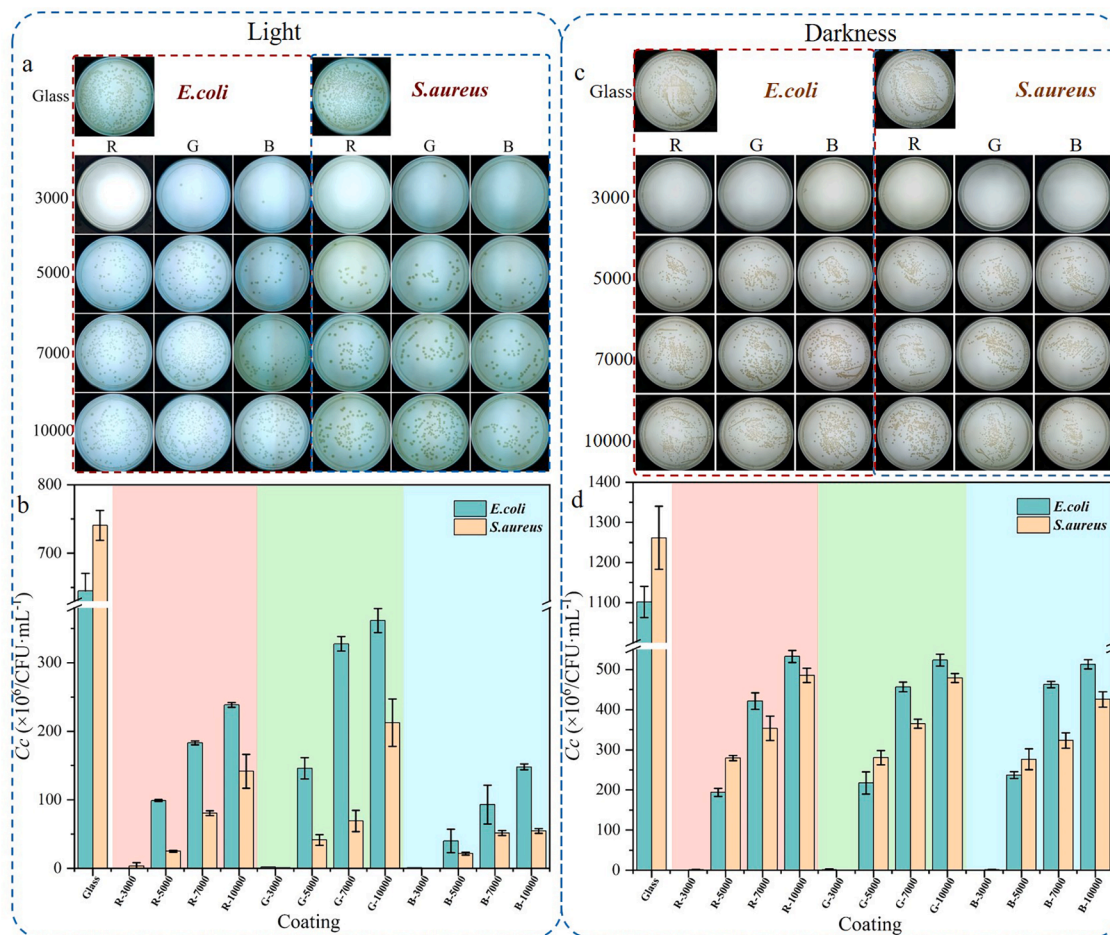


Fig. 6. (a) Photographs of bacterial colonies under 24 h continuous illumination. (b) Number of bacterial colonies under 24 h continuous illumination. (c) Photographs of bacterial colonies under 24 h continuous darkness. (d) Number of bacterial colonies under 24 h continuous darkness.

fluorescent coatings before and after healing are shown in S8-S9. Post-healing tensile tests revealed diminished mechanical performance across all coatings. Young's moduli of healed samples decreased by nearly 50 % compared to pristine coatings. This decline arises from irreversible scission of siloxane backbone bonds during cutting, which are not dynamically reversible. While imine bond reformation drives self-healing, the permanent loss of siloxane integrity limits full mechanical recovery. Despite this, the observed healing efficiency remains remarkable for a dynamically crosslinked system. The self-healing efficiency, calculated based on the tensile fracture strength before cutting and after healing, is presented in Table 1. In terms of self-healing efficiency, the red fluorescent coating is slightly lower than the other two groups. This is speculated to be due to the fact that the red fluorescent molecule possesses only two terminal aldehyde groups, whereas the other two groups each have four terminal aldehyde groups.

3.7. Antibacterial adhesion performance

The anti-adhesion efficacy of the coatings against *Escherichia coli* (*E. coli*) and *Staphylococcus aureus* (*S. aureus*) was evaluated under 24 h

Table 2
Antibacterial rate of coatings.

Coating	Light		Darkness	
	Antibacterial rate (%)		Antibacterial rate (%)	
	<i>E.coli</i>	<i>S.aureus</i>	<i>E.coli</i>	<i>S.aureus</i>
R-3000	100	99.59	100.00	99.92
R-5000	84.65	96.62	82.39	77.84
R-7000	71.63	89.19	61.73	71.98
R-10000	63.1	80.95	51.66	61.51
G-3000	99.69	99.86	99.86	100.00
G-5000	77.36	94.46	80.25	77.76
G-7000	49.3	90.68	58.56	71.07
G-10000	44.03	71.35	52.47	62.03
B-3000	99.84	99.99	100.00	99.92
B-5000	93.80	97.16	78.48	78.08
B-7000	85.58	92.97	58.01	74.36
B-10000	77.05	92.57	53.43	66.27

of continuous illumination and dark conditions. As shown in Fig. 6 and Table 2, the fluorescent coatings exhibited significant anti-bioadhesion performance compared to control glass substrates, with variations

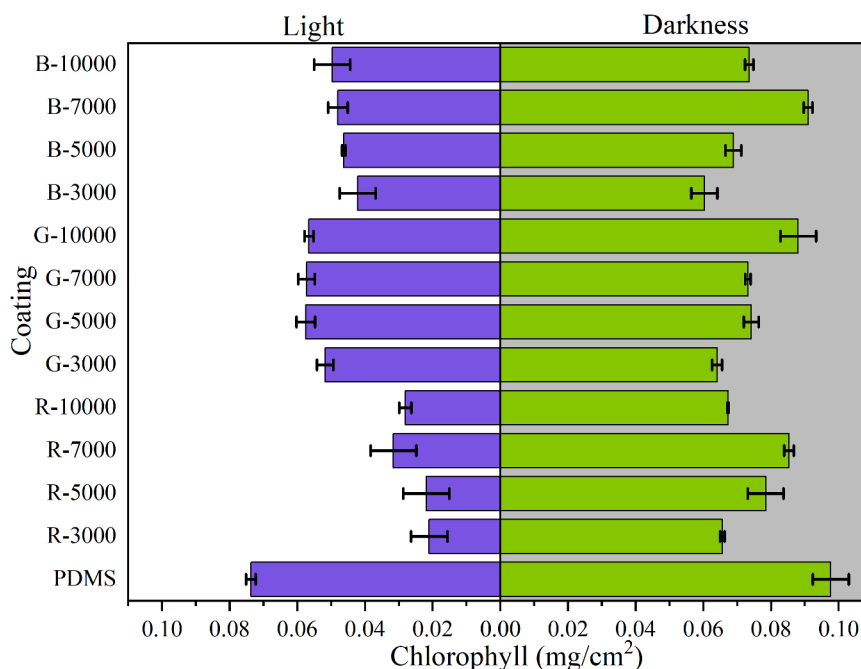


Fig. 7. Chlorophyll content of coating surface under light and dark conditions.

Table 3

Anti-algae attachment ratio of coatings.

Coating	Anti-algae rate (%)		Coating	Anti-algae rate (%)		Coating	Anti-algae rate (%)	
	Light	Darkness		Light	Darkness		Light	Darkness
R-3000	71.49	32.81	G-3000	29.77	34.42	B-3000	42.76	38.27
R-5000	70.31	19.71	G-5000	22.02	24.08	B-5000	37.19	29.46
R-7000	57.16	12.63	G-7000	22.32	25.11	B-7000	34.87	6.89
R-10000	61.94	31.13	G-10000	23.27	9.86	B-10000	32.56	24.69

observed among coatings emitting different wavelengths. Notably, blue-emitting coatings demonstrated the highest antibacterial efficacy, likely due to the inherent bactericidal properties of blue light [36–39]. Notably, the coatings also maintained considerable anti-adhesion activity in the dark, although the efficacy was lower than that under illuminated conditions. It is hypothesized that the blue light component in the illumination inhibits bacterial activity, whereas in darkness, bacterial proliferation and adhesion capabilities are enhanced.

Intriguingly, coatings prepared with lower molecular weight PDMS monomers exhibited superior antibacterial performance. For instance, the coating with a monomer molecular weight of 3000 achieved nearly complete bacterial resistance. This phenomenon is attributed to differences in surface-exposed aldehyde groups, which possess intrinsic contact-killing activity against bacteria [40,41]. During the Schiff base reaction between terminal amines of PDMS and aldehyde-containing fluorophores, a fraction of aldehyde groups remained unreacted and accessible on the coating surface. Lower molecular weight PDMS monomers contain a higher density of terminal amine groups per unit weight, necessitating a greater amount of aldehyde-functionalized crosslinkers. Consequently, coatings derived from smaller monomers retained more exposed aldehyde groups, thereby enhancing their antibacterial potency.

3.8. Anti-Diatom adhesion performance

Navicula tenera (*N. tenera*), a well-known benthic diatom, commonly adheres to surfaces and contributes to biofouling. The anti-diatom adhesion performance of the coatings was evaluated by measuring

chlorophyll-a content extracted from attached diatoms. As shown in Fig. 7 and Table 3, coatings exposed to continuous light exhibited significantly lower diatom adhesion compared to those under complete darkness, confirming that light influences the physiological behavior of *N. tenera*. Furthermore, fluorescent coatings demonstrated superior anti-algal adhesion efficacy relative to conventional PDMS coatings under illuminated conditions. Notably, red-emitting coatings outperformed those emitting other wavelengths, likely due to enhanced absorption of red light by the diatoms, which may disrupt their attachment mechanisms. This wavelength-dependent inhibition was absent in darkness, underscoring the role of specific light wavelengths in suppressing diatom colonization.

3.9. pH-Responsive self-renewal

The imine bonds within the coatings are acid-labile, undergoing hydrolysis under acidic conditions to enable pH-triggered dissolution and self-renewal. Bacterial metabolic activity generates acidic micro-environments, with biofilm pH levels reaching as low as 2 [42]. This property is harnessed to establish an antifouling mechanism [43], wherein localized acidification by bacterial biofilms induces partial dissolution of the coating surface, thereby inhibiting bacterial adhesion and biofilm proliferation. This self-renewing antifouling behavior is accompanied by the release of fluorescent molecules, allowing dissolution dynamics to be quantified via absorbance measurements of the released fluorophores. For instance, the green-emitting fluorophore TPB exhibits a characteristic absorption peak at 239 nm. Thus, the dissolution extent of green-emitting coatings was assessed by monitoring

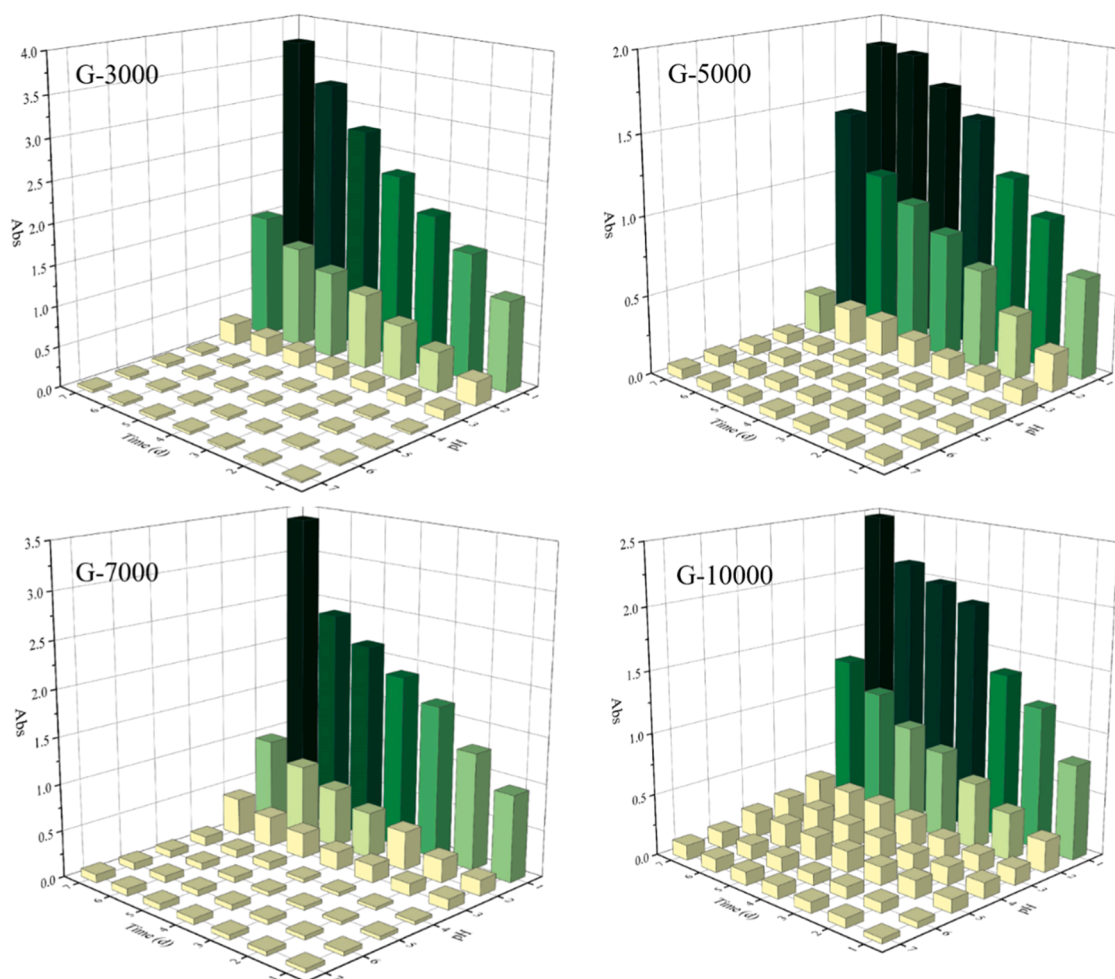


Fig. 8. TPB absorbance released from green coatings.

absorbance at 239 nm (Fig. 8). All four green fluorescent coatings (varying in PDMS molecular weight) displayed similar pH-dependent release profiles. Minimal TPB release occurred at pH 4–7. Release increased moderately at pH 3 and surged sharply at pH 2, peaking at pH 1. Other coatings are similar to the G-3000. This demonstrates that the dissolution of the coating is pH-dependent and requires a specific pH threshold to be reached. As shown in the figure, this condition corresponds to a pH of 4. This pH-responsive release aligns closely with the acidic conditions generated by bacterial biofilms, confirming the coatings' pH-activated self-cleaning and antifouling capabilities.

4. Conclusions

In summary, three aldehyde-functionalized fluorophores with distinct emission wavelengths—red-emitting BTB, green-emitting TPB, and blue-emitting SPB were synthesized, and their photophysical behaviors were systematically investigated. These fluorophores served as crosslinkers to fabricate a series of fluorescent antifouling coatings by reacting with amine-terminated polydimethylsiloxane (PDMS) of varying molecular weights (3000, 5000, 7000, and 10000). The covalent bonding mechanism between the fluorophores and PDMS, along with the fluorescence properties of the coatings, were thoroughly characterized. The self-healing behavior and associated mechanical property changes (pre- and post-healing) were elucidated, revealing the critical role of dynamic imine bonds in enabling autonomous repair.

Antibacterial efficacy was determined to arise from a synergistic interplay between the fluorescence wavelength and surface-exposed aldehyde groups, which exhibit contact-killing activity. Additionally,

the coatings inhibited diatom adhesion by disrupting algal photosynthesis through wavelength-specific light emission. Notably, the acid-labile nature of the imine bonds endowed the coatings with pH-responsive self-cleaning capabilities, effectively suppressing biofilm formation under acidic conditions generated by bacterial metabolism. Thus, the synergistic integration of fluorescent antifouling, silicone-derived low surface energy, and pH-responsive self-renewal endows these coatings with exceptional antifouling performance.

CRediT authorship contribution statement

Haicheng Yang: Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yuhong Qi:** Writing – review & editing, Funding acquisition, Formal analysis, Conceptualization. **Zhanping Zhang:** Writing – review & editing, Project administration, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.surf.2026.108527](https://doi.org/10.1016/j.surf.2026.108527).

Data availability

Data will be made available on request.

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